

Advances in Ozone Measurement Traceability

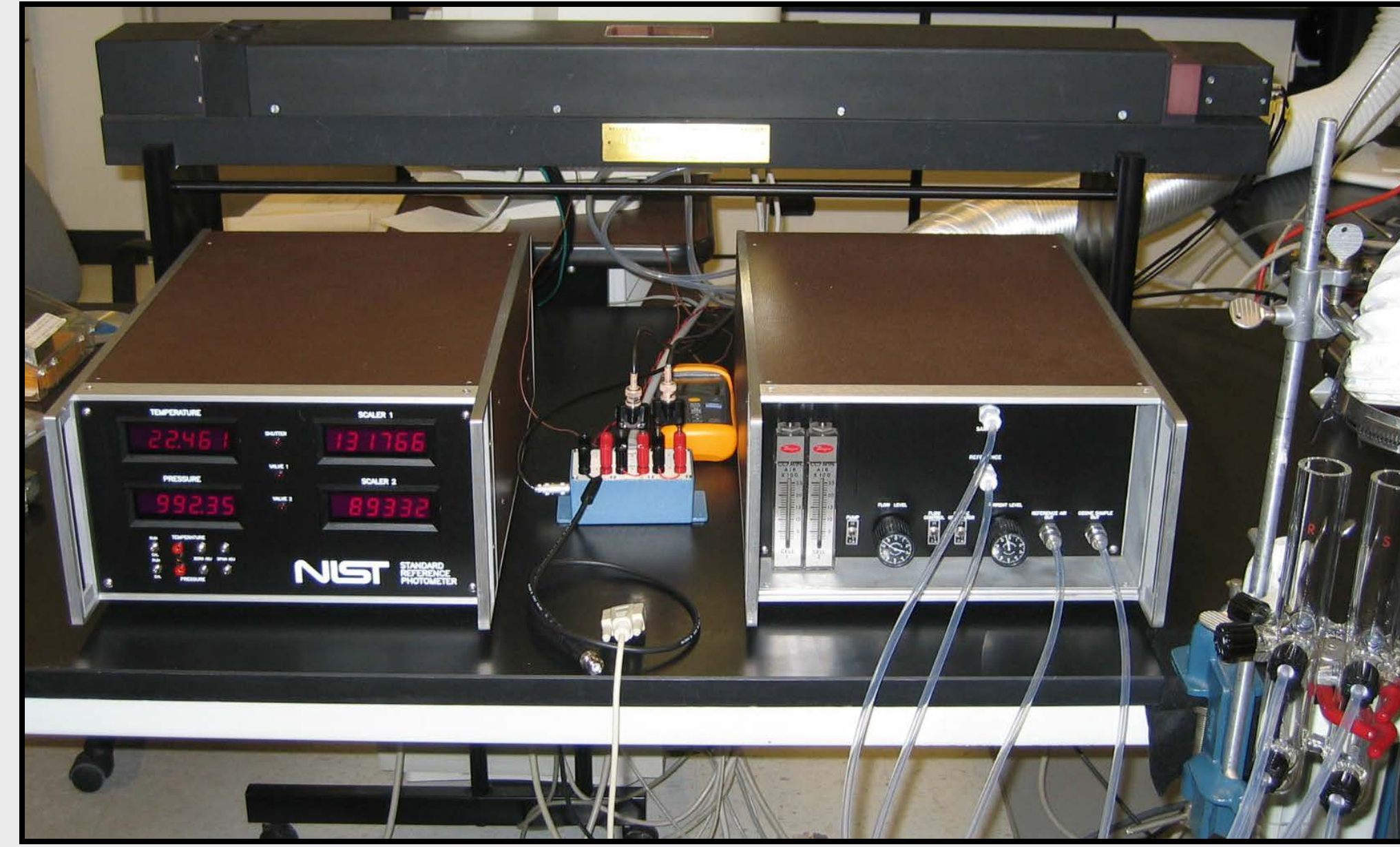
Abstract # 2010-A-835-AWMA

James E. Norris,¹ Robert Wielgosz,² Joëlle Viallon,² Franklin R. Guenther¹
¹NIST Analytical Chemistry Division, ²BIPM Chemistry Section, Sevres, France

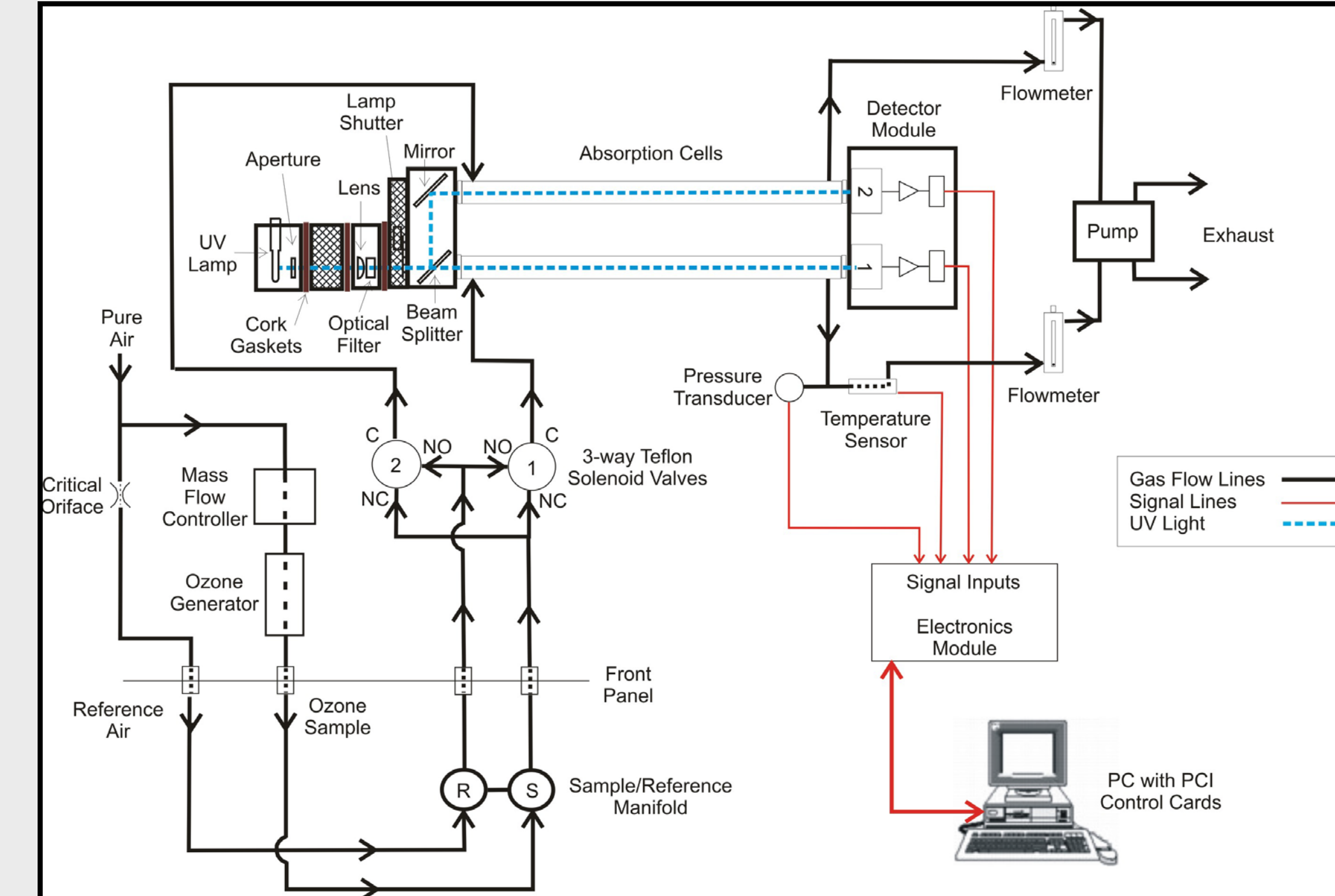
NIST Ozone Standards Program

- The NIST Standard Reference Photometer (SRP) was originally developed by NIST and the USEPA to provide Traceability of Ozone Measurements within the United States to NIST.
- A network of NIST SRPs were built and disseminated throughout the U.S. in USEPA Regional Laboratories to provide a reliable Reference Standard for U.S. Ambient Ozone Measurements under U.S. EPA regulation.
- Interest from around the World prompted NIST to provide the NIST SRP to an increasing number of Laboratories Worldwide.
- Provides ozone traceability for U.S. Industry, Academia, and other Government Agencies.
- Reference Standard for the WMO Global Atmospheric Watch Program.
- Collaboration with BIPM to establish International Ozone Traceability System.
- Continue to provide NIST SRPs to Government Laboratories around the World.

NIST Standard Reference Photometer



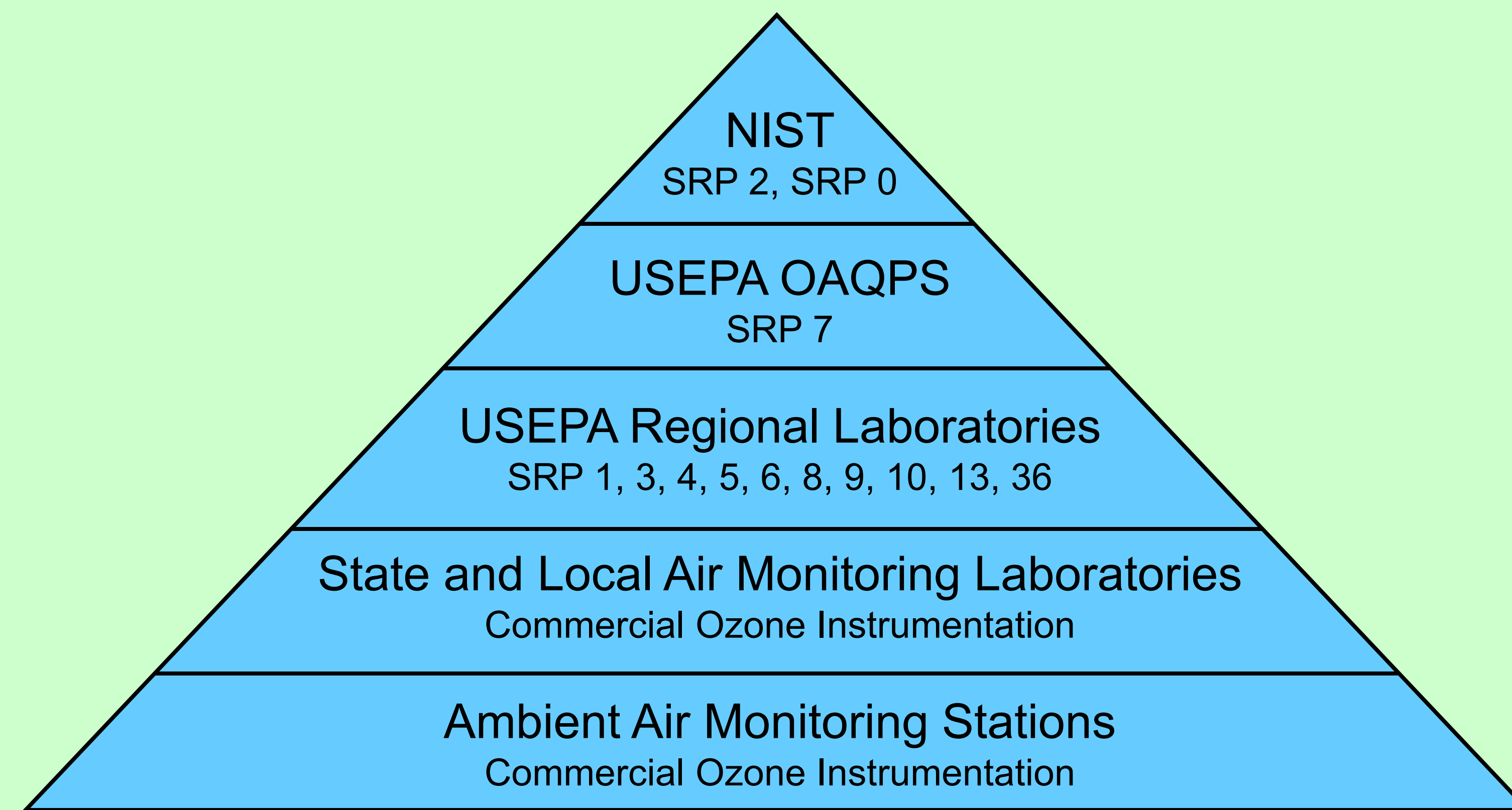
NIST SRP Schematic



NIST SRPs Worldwide



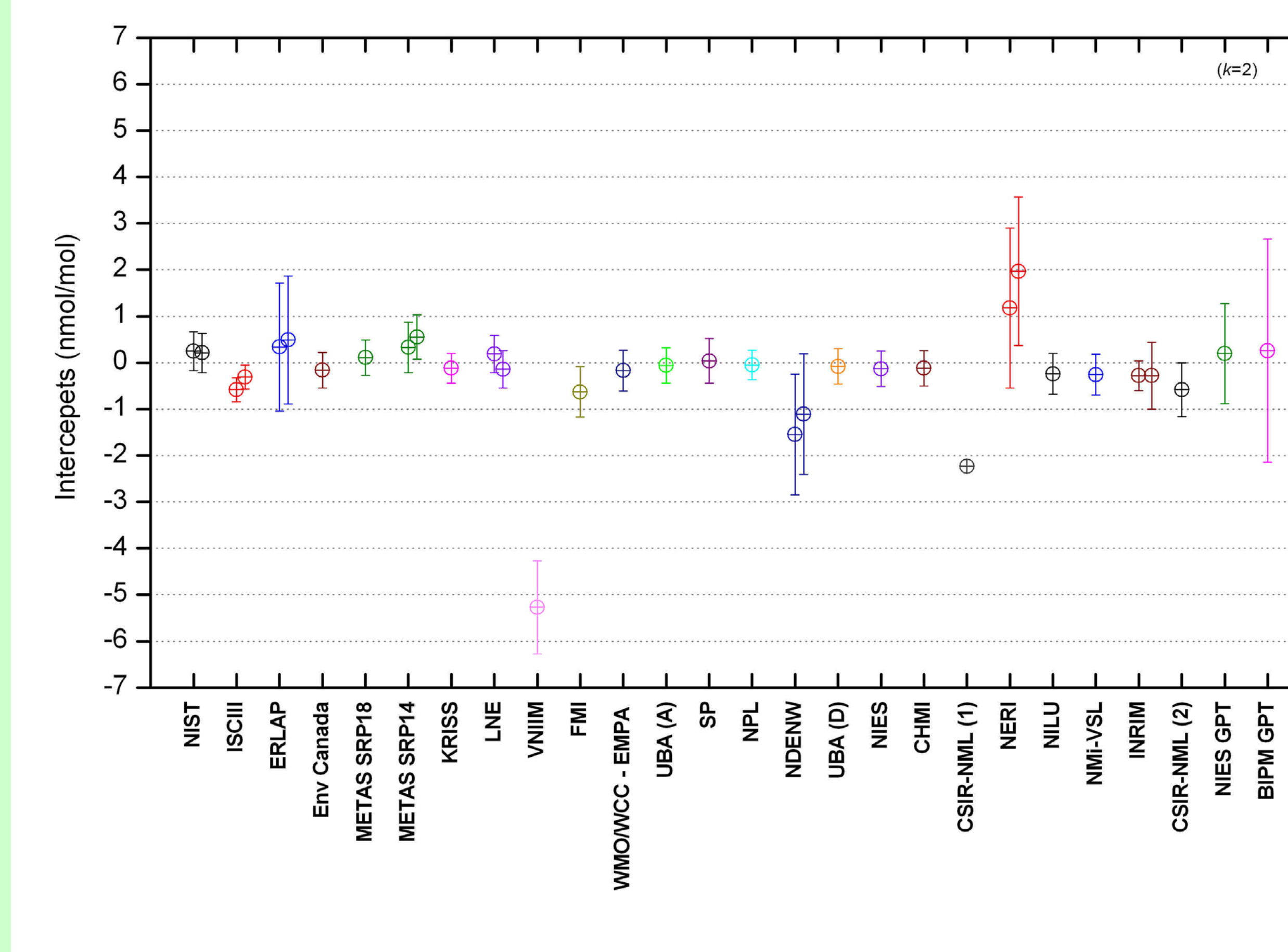
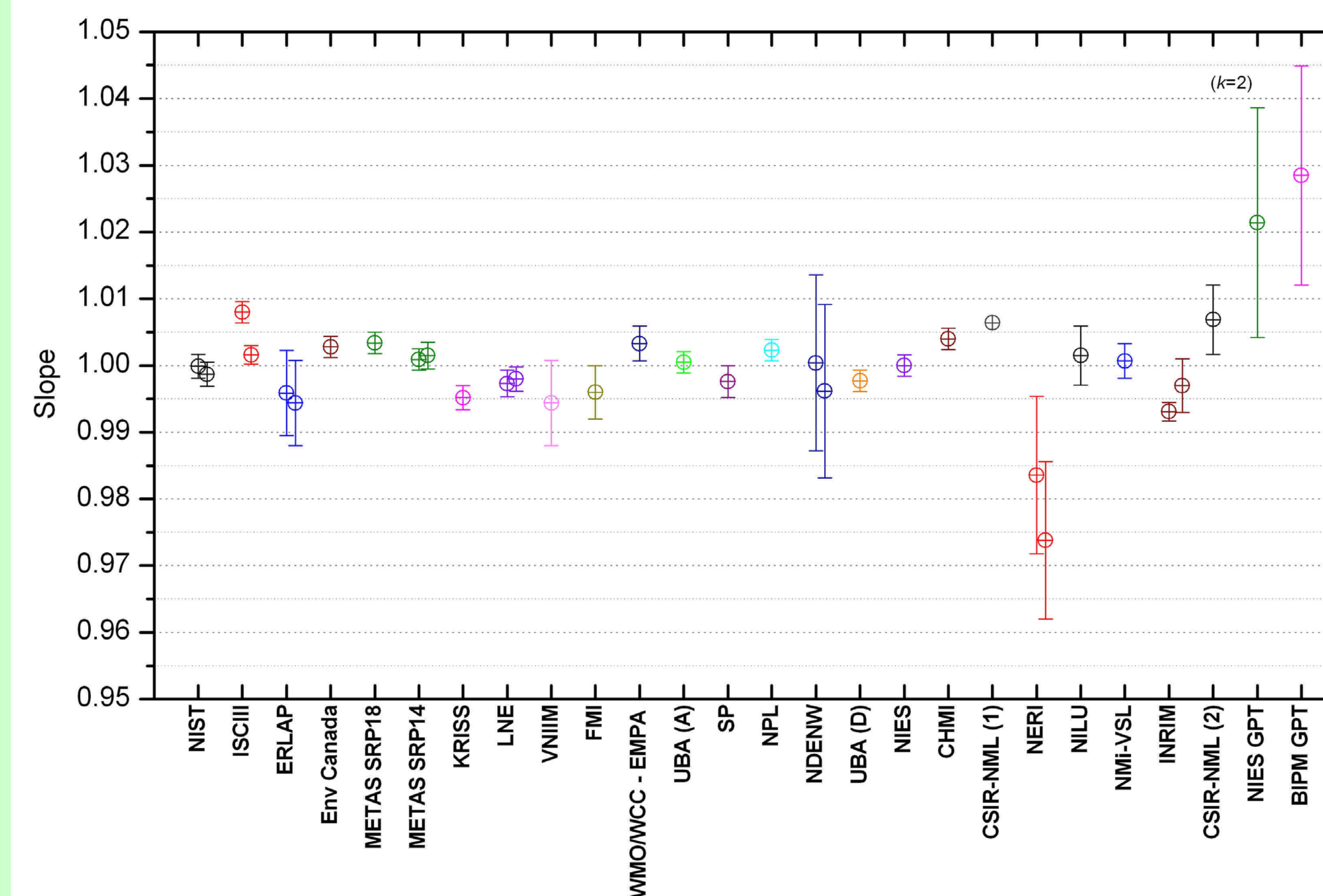
Traceability of Ambient Ozone Measurements in the United States



- Annual inter-comparisons of USEPA OAQPS SRP 7 against NIST SRP 2 (or SRP 0 as a transfer).
- Annual inter-comparisons of all USEPA Regional Laboratory SRPs against USEPA OAQPS SRP 7.
- Annual inter-comparisons of State and Local Air Monitoring Agencies against USEPA Regional Laboratory SRPs.
- Calibrations of Ambient Air Monitors by State and Local Air Monitoring Agencies.

CCQM-P28 Pilot Study, ozone at ambient level

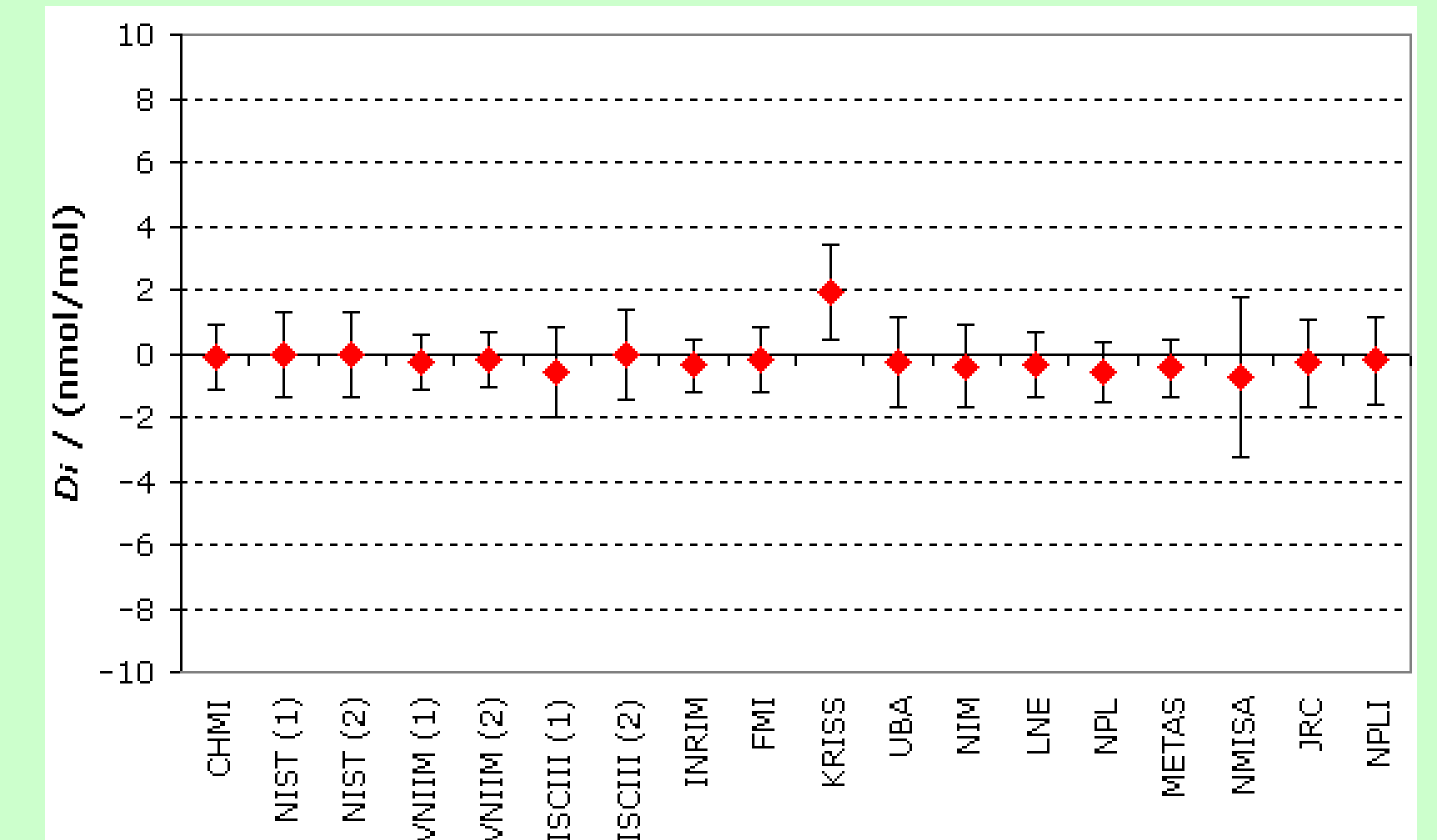
- Inter-comparisons against BIPM SRP 27 (reference standard).
- 23 participants from 20 Countries.
- Protocol A – Direct Comparison of National Standard vs. Reference Standard.
- Protocol B – Transfer Standard Comparison vs. National and Reference Standard.
- UV Photometric Instruments of various types.
- 2 Gas Phase Titration Instruments (GPT)
- Unresolved 2-3% difference between UV Photometric and GPT measurements!



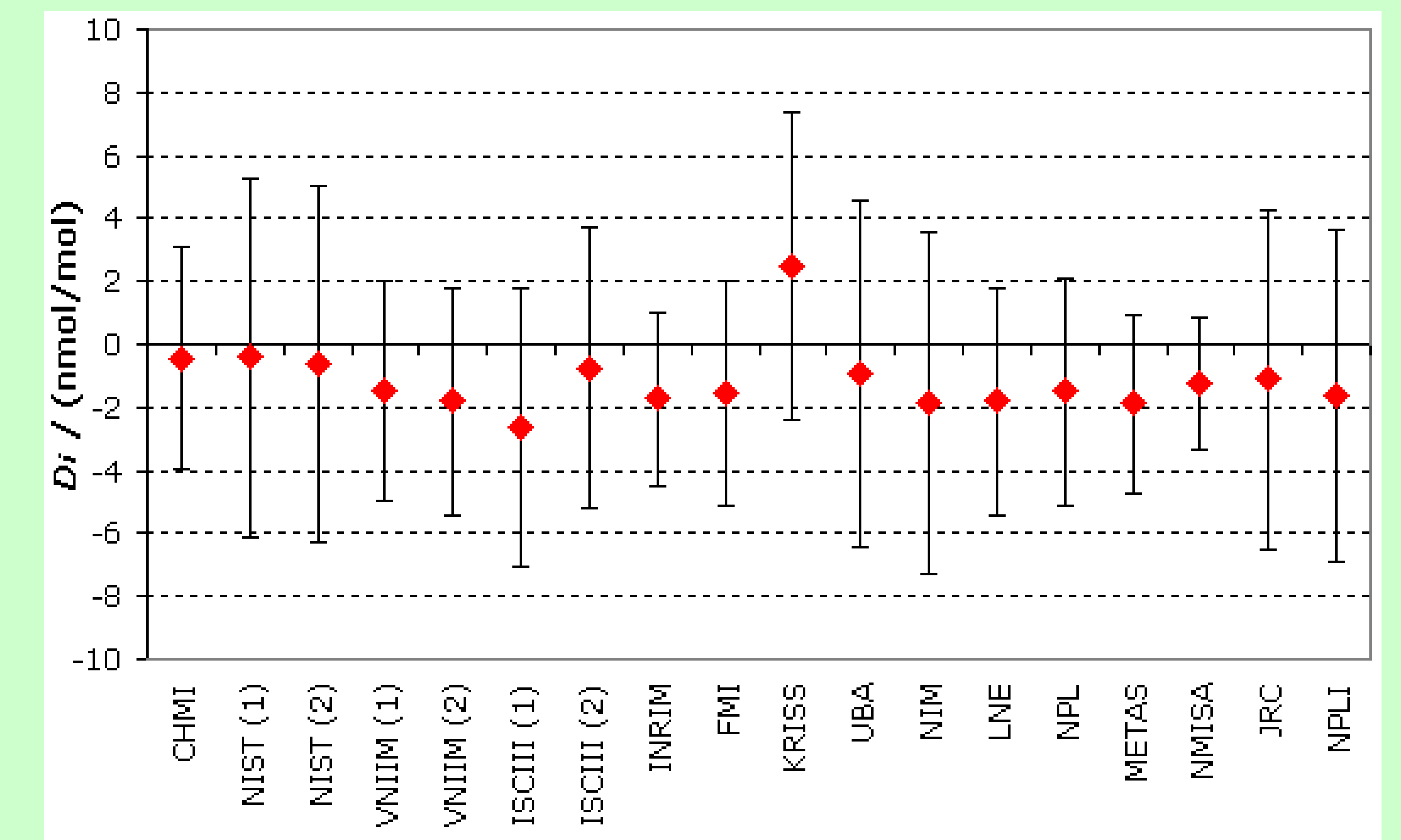
BIPM.QM-K1 Key Comparison, ozone at ambient level

- Voluntary participation every 2 years.
- Results at 80 nmol/mol and 420 nmol/mol against BIPM SRP 27 (reference standard).
- 15 Countries participated in 2007-2008 round.
- Protocol A and B followed as in CCQM-P28 Pilot Study.
- All UV Photometric Instruments (mostly NIST SRPs).

Difference from Reference Value at 80 nmol/mol



Difference from Reference Value at 420 nmol/mol



International Traceability of Ambient Ozone Measurements

- Meter Convention – Diplomatic Treaty between 54 Nations.
- CIPM – International Committee for Weights and Measures.
- BIPM - International Bureau of Weights and Measures (Reference Laboratory) under CIPM.
- NMIs - National Measurement Institutes of Member States and Associate Members States of the Meter Convention.
- NMIs or Designated Laboratories participate in BIPM.QM-K1 Key Comparison, ozone at ambient level.
- Traceability within each Country provided by NMI or Designated Laboratory.
- Traceability can also be obtained through Regional Metrology Organization Key Comparisons.

